

CS 280 Examination

Discrete Mathematics and Combinatorics – Answer Key

1. **(B)** A bipartite graph has chromatic number at most 2.
 A connected bipartite graph has chromatic number exactly 2 (color each partition with a different color). However, a disconnected bipartite graph with isolated vertices would have chromatic number 1. Thus, bipartite graphs have $\chi(G) \leq 2$.
2. **(C)** $\binom{n+k-1}{k-1}$, the number of non-negative integer solutions to $x_1 + \dots + x_k = n$.
 The expansion of $(1-x)^{-k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n$. This counts the “stars and bars” problem: distributing n identical objects into k distinct bins.
3. **(A)** $\exists x \forall y \neg P(x, y)$.
 To negate a quantified statement, flip each quantifier and negate the predicate. The negation of “for all x there exists y such that $P(x, y)$ ” is “there exists x such that for all y , not $P(x, y)$ ”.
4. **(D)** In any sequence of $n^2 + 1$ distinct real numbers, there exists a strictly increasing or strictly decreasing subsequence of length $n + 1$.
 This is the Erdős-Szekeres theorem, a sophisticated application of the Pigeonhole Principle. Option (A) is correct in spirit but doesn’t explicitly state “at least”, though it’s commonly cited as the standard pigeonhole example.
5. *Sample answer:*
If direction: Suppose G is bipartite with partitions A and B . Any cycle alternates between A and B , so it must have even length. Thus, no odd cycles exist.
Only if direction: Suppose G has no odd cycles. Pick any vertex v and perform BFS. Color vertices at even distance from v with one color, and odd distance with another. If there’s an edge within the same color class, it would create an odd cycle (contradiction). Thus G is bipartite.
Application to K_n : The complete graph K_n contains triangles (3-cycles) for $n \geq 3$, which are odd. Therefore K_n is not bipartite for $n \geq 3$. For $n \leq 2$, there are no cycles at all, so K_n is trivially bipartite.
6. *Sample answer:*

- **Permutation:** An ordered arrangement of objects. The number of permutations of n objects taken r at a time is $P(n, r) = \frac{n!}{(n-r)!}$.
- **Combination:** An unordered selection of objects. The number of combinations is $C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$.
- **Inclusion-Exclusion:** Extends these by counting unions of sets. For sets $A_1, \dots, A_n$:

$$|A_1 \cup \dots \cup A_n| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \dots$$

This handles overcounting in complex scenarios like derangements or surjective functions.

7. *Sample answer:*
Handshaking Lemma: In any graph $G = (V, E)$, $\sum_{v \in V} \deg(v) = 2|E|$.

Proof: Each edge contributes 2 to the total degree sum (one for each endpoint). Therefore, the sum of all degrees equals twice the number of edges.

Corollary: The number of vertices with odd degree is even. This follows because the sum of all degrees is even ($2|E|$), and if an odd number of vertices had odd degree, the total sum would be odd (contradiction).

8. *Sample answer:*

- **Weak induction:** To prove $P(n)$, assume $P(k)$ and prove $P(k+1)$.
- **Strong induction:** To prove $P(n)$, assume $P(0), P(1), \dots, P(k)$ all hold, and prove $P(k+1)$.

Strong induction is more natural when the recursive structure depends on multiple previous cases.

Example: Proving every integer $n \geq 2$ can be factored into primes. In the inductive step for $n = k+1$: if $k+1$ is prime, done. Otherwise $k+1 = ab$ where $2 \leq a, b < k+1$. By strong induction, both a and b have prime factorizations, so $k+1$ does too. Weak induction would not easily give us information about both a and b .